

Course Code	Name of the Course			
216U06I306	Foundational Course in Indian Knowledge Systems			
Teaching Scheme	TH	P	TUT	Total
	02	--	--	02
Credits Assigned	02	--	--	02
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	--	50	--	50
				100

Course Objectives:

1. to introduce students to the rich diversity of Indian knowledge systems
2. to introduce the life and works of important figures in the respective domains
3. to explore the underlying philosophical and cultural ethos that distinguishes Indian Knowledge Systems
4. to emphasise continuity of the tradition into modern times, wherever applicable.

Course Outcomes

At the end of successful completion of the course the student will

- CO1. have a clear understanding of the different domains of Indian Knowledge Systems
 CO2. have become aware of the contribution of great figures in the respective fields
 CO3. have an understanding of how culture impacts creation of knowledge
 CO4. learn to investigate correlations and synthesis leading to development of any knowledge system

Detailed Syllabus

Module No.	Unit No.	Topics	Hours per topic	Hours per Unit
1		Sources of Indian Knowledge Systems		4
	1.1	IKS - Concept, scope, relevance to our world today.	1	
	1.2	Textual sources, historical accounts, archaeological evidence, inscriptions, coins etc	3	
2		Why study IKS?		2
	2.1	Importance of the IKS, its interconnections and relevance to the modern fields of science.	2	
3		Yoga: Basic Practices and Philosophy		6
	2.1	Maharshi Patanjali, Swami Satyananda Saraswati, B K S Iyengar, Swami Kuvalayananda, Sri Yogendra	2	
	2.2	Body loosening exercises Importance of breath, developing concentration Yoga for mind-body wellness	4	
4		Genres of Ancient Literature		6
	4.1	Religious: Vedic texts, Buddhist and Jain texts;	2	
	4.2	Epics, Puranas, Sangam literature	2	
	4.3	Poetry, Mathematics, and Scientific Literature	2	
5		Leadership and Ethical Values		6
	5.1	Selections from Shantiparva of Mahabharata, Arthashastra, Panchatantra, Hitopadesha, Jataka tales, Bhagavadgita, Dhammapada and Thirukkural: Discussions on ethical values	3	
	5.2	Leadership qualities as reflected through ancient Indian literature: Lessons for modern leadership challenges	3	
6	6.1	Art: Sculpture(iconography) and Paintings	3	
		Iconography: Ellora (Buddhist and Jain) and Hampi (Hindu)		
		Paintings: Ajanta(Buddhist), Ellora(Jain), Brihadeshvar Temple-Thanjavur.(Hindu)		
	6.2	Architecture: Rock-cut caves and Temple Architecture	3	
		Rock-cut caves: Kanheri, Elephanta, Ellora (any two sites can be used for detailed discussion)		

		Temple architecture: Pattadakal, Konark Temple, Jagannatha Temple-Puri, Bodh Gaya, Dilwara Temple-Mount Abu (any two sites can be used for detailed discussion)		
7		Ancient Indian Mathematics		6
	7.1	Shulba Sutras, Bakshali Manuscript	2	
	7.2	Aryabhatiya: place value system, approximation of the value of π , geometry	2	
	7.3	Bhaskaracharya: different approach to teaching mathematics	2	
8		Ancient Indian Astronomy		6
	8.1	Indian calendar system: Sayana-nirayana calendar, Panchanga	3	
	8.2	Spherical trigonometry, Eclipse computation	3	
9		Ancient Indian Agriculture		6
	9.1	General management of Agriculture and Farming Operations	3	
	9.2	Cattle Management, Weather predictions	3	
10		Trade and Commerce		6
	10.1	Silk route, Uttarapatha and Dakshinapatha, Maritime route	3	
	10.2	Barter system, Numismatics	3	
11		Ancient Indian Society		6
	11.1	Law and Justice	3	
	11.2	Marriage Laws, Inheritance	3	
12		Chemistry and Metallurgy		6
	12.1	Multiple sources such as archaeological artifacts, temple icons,	1	
	12.2	Metals and beads	2	
	12.3	Chemistry of dyes, Colouring materials	2	
	12.4	Paintings and Painting materials	1	
Total Hours				30*

* The first two modules remain the core and other modules can be selected (any 4 modules from module 3 to module 12) by the college depending upon the availability of the teachers making it to a 30hrs course.

Recommended Books:

Text Book on IKS:

1. Mahadevan B., Bhat Vinayak Rajat, Nagendra Pavana R. N. Introduction to Indian Knowledge System: concepts and Applications, PHI Learning Pvt. Ltd. 2022

Reference Books:

1. Amma Sarasvati T. A., Geometry in Ancient and Medieval India, MLBD, Delhi, 1st ed. 1999, reprint 2007
2. Acharya, P. K., Indian Architecture According to ManasaraShilapshastra, Oxford University Press 1927
3. Altekar, A.S., Education in Ancient India, Gyan Books, 2010
4. Appleton Naomi, Jataka Stories in Theravada Buddhism: Narrating the Bodhisatta Path, Routledge Publication, New York 2016
5. Bhattacharyya, T. , Study of Vastuvidya or Canon of Indian Architecture, Patna 1976
6. Bose, N. K., Orissan temple Temple Architecture (Vastushastra) [With Sanskrit text and English translation), Bharatiya Kala Prakashana, Delhi 20017
7. Chatterjee, Satischandra & Datta, Dharendra Mohan. An introduction to Indian Philosophy, Rupa Publications India Pvt. Ltd., New Delhi, 7th edition, 1968
8. Clark Walter Eugene, The Aryabhatiya of Aryabhata- An Ancient Indian Work On Mathematics and Astronomy, Delta Book World, India, 2021
9. Coomaraswamy, Ananda K. Early Indian Architecture: Cities and City-Gates, Munshiram Manoharlal Publishers, 2002
10. D M Bose, S N Sen and B V Subbarayappa, eds; A Concise History of Science in India, INSA; 2009
11. Datta Bibhutibhushan & Singh Avadhesh Narayan, History of Hindu Mathematics, 1935, repr. Bharatiya Kala Prakashan, Delhi, 2004
12. Datta Bibhutibhushan, Ancient Hindu Geometry: The Science of the Śulba, 1932, reprint. Cosmo Publications, New Delhi, 1993
13. Deglurkar, G. B, Temple Architecture and Sculpture of Maharashtra, Nagpur University, Nagpur 1974
14. Dehejia, Vidya, Early Buddhist Rock Temples A Chronological Study, London, 1972
15. Dehejia, Vidya, Early Stone Temples of Orissa, Vikas Publishing House, Delhi 1979
16. Divakaran P. P., The Mathematics of India: Concepts, Methods, Connections, Hindustan Book Agency, 2018
17. Dr. Mishra Shiv Shekhar, Fine Arts & Technical Sciences in Ancient India with special reference to Someśvara's Mānasollāsa; Krishnadas Academy, Varanasi 1982
18. Ed. and Trs. Majumdar Girija Prasanna, Banerji Sures Chandra, Kriśi-Parasara, Asiatic Society, Kolkata, 1960
19. Ed. Tr. Kangale, R. P, Kautiliya Arthashastra, University of Bombay, Bombay, 1960
20. Gupta, Swarajya Prakash, Asthana Shashi, Elements of Indian Art: Including Temple Architecture, Iconography & Iconometry, Indraprastha Museum of Art and Archeology, 2007
21. Kane P.V., History of Sanskrit Poetics, Motilal Banarasidass, New Delhi, 4th edition, 1971

22. Larson, G. J. (Ed.) and Bhattacharya, R. (Ed.) , Encyclopaedia of Indian Philosophies: Yoga: India's Philosophy of Meditation, Vol. XII, Motilal Banarasidas Publishers Pvt. Ltd., Delhi, 1st edi., 2008
23. Paranjpe Kalpana, Ancient Indian insights and Modern Science: A Rare Book, Bhandarkar Oriental Research Institute, Pune, 2022
24. Radhakrishnan, S., The Principal Upanisads, Oxford University Press, Delhi, 1992
25. Rahman A., Alvi M. AKhan .S A., Ghorl, Murthy Samba K. V., Science and Technology in Medieval India - A Bibliography of Source Materials in Sanskrit, Arabic and Persian, 1982
26. Rao Balachandra S., Indian Astronomy – An Introduction, Universities Press (India) Limited, Hyderabad, 2000